

TalkTrace Complete System Architecture & Database Spec

Combined Unified Document: Frontend (Web Client) + Backend (Node & Python Engine)

SECTION 1: SYSTEM OVERVIEW & REPOSITORY STRUCTURE

TalkTrace is an enterprise-grade, zero-trust meeting intelligence application. It provides real-time audio capture, local Whisper transcription, Pyannote speaker diarization, Google Translate language detection, client-side sentence embedding vector RAG search, LLM executive minutes extraction, task management, and Google/Outlook calendar auto-joins.

SECTION 2: FRONTEND WEB ARCHITECTURE (talktrace-web)

Frontend Component Specs & State Architecture

Overview of React UI components, security layers, client-side vector search engines, and auth contexts.

No	Component / Module	Type	Responsibilities	Security / Tech Stack
1	Dashboard.js	React View	Core workspace UI, loading stepp...	Tailwind, Lucide, jsPDF
2	GlobalSearch.js	React Search	128-d sentence vector embeddings...	IndexedDB, Math Vector
3	CalendarSync.js	React Module	Google & Outlook calendar meetin...	REST API, Token Auth
4	TaskManager.js	React Module	AI action item extraction board,...	Local State, Exporters
5	Analytics.js	React View	Speaker talk-time charts, WPM sp...	Canvas, Math Metrics
6	AuthContext.js	React Context	Firebase authentication, email v...	Firebase Auth SDK
7	Login.js / Signup.js	React Views	User registration with email ver...	Firebase Auth

SECTION 3: BACKEND SERVICE & ENGINE ARCHITECTURE

Backend Modules, Pipelines & AI Engines

Server REST endpoints, zero-trust vault storage, Python diarization, local Ollama NLP, and calendar sync.

No	Backend Module	Tech / Runtime	Description & Functions	Key Security Layer
1	server.js	Node.js Express	REST API server, RSA/AES key exc...	Bearer Token Auth
2	pipeline_manager.js	Node.js Stream	FFmpeg ingestion, temp audio cle...	Stream Encryption
3	storage_enc.js	Node.js Crypto	Local zero-trust storage vault, ...	AES-256 Cryptography
4	database.js	SQLite3	Database layer for meetings, key...	Encrypted Blob Vault
5	diarize.py	Python 3.10	WhisperX transcription, Pyannote...	Torch CUDA/CPU int8
6	nlp_local.js	Node.js / Ollama	Executes Llama 3.2 / Mistral for...	Local Airgapped LLM
7	calendar_sync.js	Node.js	Google Calendar & Outlook Graph ...	OAuth Tokens
8	recall.js	Node.js Axios	Recall.ai bot launcher, meeting ...	API Token Vault

SECTION 4: COMPLETE DATABASE SCHEMA SPECIFICATION

TABLE 1: users (System User Accounts)

Stores core account information for system users across all roles. Manages unique login credentials, email verification status, and security parameters.

No	Column Name	Data Type	Key / Constraint	Description
1	uid	VARCHAR(128)	PK	Unique Firebase / Auth User Id...
2	name	VARCHAR(100)	NOT NULL	Full Name of User
3	email	VARCHAR(150)	UNIQUE, NOT NULL	Login Email Address
4	emailVerified	BOOLEAN	DEFAULT FALSE	Email Verification Confirmatio...
5	role	ENUM(admin,user)	DEFAULT user	Access Control Role
6	createdAt	TIMESTAMP	DEFAULT NOW()	Account Creation Date

TABLE 2: meetings (Meeting Sessions & Audio Meta)

Holds metadata for recorded meeting sessions, including language auto-detection, audio duration, processing lifecycle state, and owner association.

No	Column Name	Data Type	Key / Constraint	Description
1	id	VARCHAR(128)	PK	Unique Meeting UUID
2	user_id	VARCHAR(128)	FK (users.uid)	Owner User Identifier
3	bot_id	VARCHAR(128)	NULLABLE	Recall.ai Note-Taker Bot ID
4	status	VARCHAR(50)	DEFAULT processing	Overall Status (processing, co...
5	process_state	VARCHAR(50)	DEFAULT initializing	Detailed Pipeline State (trans...
6	language	VARCHAR(10)	DEFAULT auto	Google / Whisper Verified Lang...
7	duration_seconds	INTEGER	DEFAULT 0	Audio Duration in Seconds
8	created_at	BIGINT	NOT NULL	Session Epoch Timestamp (ms)

TABLE 3: meeting_keys (Zero-Trust Cryptographic Key Vault)

Encrypted vault storing AES-256 keys and initialization vectors (IV) for local zero-trust audio and transcript encryption.

No	Column Name	Data Type	Key / Constraint	Description
1	meeting_id	VARCHAR(128)	PK, FK (meetings.id)	Associated Meeting ID
2	encrypted_key_blob	TEXT	NOT NULL	Protected Key Vault Blob (AES/...

TABLE 4: transcript_revisions (Version History & Hash Audit Log)

Tracks version history and SHA-256 cryptographic hashes for transcripts and executive summaries to guarantee audit immutability.

No	Column Name	Data Type	Key / Constraint	Description
1	id	INTEGER	PK, AUTOINCREMENT	Revision Record ID
2	meeting_id	VARCHAR(128)	FK (meetings.id)	Associated Meeting ID
3	version	INTEGER	NOT NULL	Revision Version Number (1, 2,...
4	type	ENUM(transcript, summary)	NOT NULL	Revision Artifact Type
5	content_hash	VARCHAR(64)	NOT NULL	SHA-256 Cryptographic Hash
6	file_path	VARCHAR(255)	NOT NULL	Encrypted Vault File Path
7	edited_at	BIGINT	NOT NULL	Revision Edit Timestamp (ms)

TABLE 5: speaker_profiles (Recurring Speaker Voice & Name Registry)

Stores persistent participant voice profiles and custom name mappings across meetings for automated Pyannote speaker recognition.

No	Column Name	Data Type	Key / Constraint	Description
1	id	INTEGER	PK, AUTOINCREMENT	Profile Record ID
2	user_id	VARCHAR(128)	FK (users.uid)	Owner User ID
3	speaker_label	VARCHAR(50)	NOT NULL	Pyannote Diarization Label
4	speaker_name	VARCHAR(100)	NOT NULL	Real Human Name (e.g. Abin Geo...
5	voice_signature	TEXT	NULLABLE	Dense Voice Embedding Vector
6	updated_at	BIGINT	NOT NULL	Last Update Timestamp (ms)

TABLE 6: action_items (AI Action Items & Task Manager Board)

Contains structured tasks automatically extracted by local LLMs (Ollama) from meeting transcripts, with priorities.

No	Column Name	Data Type	Key / Constraint	Description
1	id	VARCHAR(128)	PK	Action Item Unique ID
2	meeting_id	VARCHAR(128)	FK (meetings.id)	Originating Meeting ID
3	task	TEXT	NOT NULL	Action Item Task Description
4	assignee	VARCHAR(100)	DEFAULT Unassigned	Responsible Participant
5	deadline	VARCHAR(50)	DEFAULT ASAP	Targeted Due Date / Timeframe
6	priority	ENUM(High,Medium,Low)	DEFAULT Medium	Task Urgency Rating
7	completed	BOOLEAN	DEFAULT FALSE	Task Completion Status
8	confidence	FLOAT	DEFAULT 0.95	LLM Extraction Confidence Score

TABLE 7: calendar_events (Google & Outlook Synced Meetings)

Stores synced calendar events from Google Calendar and Microsoft Outlook Graph APIs for automated bot scheduling.

No	Column Name	Data Type	Key / Constraint	Description
1	id	VARCHAR(128)	PK	Calendar Event Unique ID
2	user_id	VARCHAR(128)	FK (users.uid)	Associated User Account
3	title	VARCHAR(255)	NOT NULL	Meeting Title / Subject
4	start_time	TIMESTAMP	NOT NULL	Scheduled Start DateTime
5	meeting_link	TEXT	NOT NULL	Google Meet / Teams / Zoom URL
6	source	ENUM(Google, Outlook)	NOT NULL	Calendar Source API
7	bot_scheduled	BOOLEAN	DEFAULT FALSE	Auto-Join Bot Scheduled Flag

TABLE 8: search_history (Natural Language Vector RAG Query Logs)

Stores natural language search queries and 128-dimensional sentence embedding vector logs for client-side search audit.

No	Column Name	Data Type	Key / Constraint	Description
1	id	INTEGER	PK, AUTOINCREMENT	Log ID
2	user_id	VARCHAR(128)	FK (users.uid)	Query User ID
3	query	TEXT	NOT NULL	Natural Language Search String
4	created_at	BIGINT	NOT NULL	Search Timestamp (ms)